Solving Linear Equations

3-L U Decomposition

Why L U Decomposition

- Gauss elimination is designed to solve systems of linear algebraic equations.
- Although it certainly represents a sound way to solve such systems, it becomes **inefficient when solving equations with the same coefficients [A], but with different right hand-side constants** (the b's).
- LU decomposition methods separate the time-consuming elimination of the matrix [A] from the manipulations of the right-hand side {B}. Thus, once [A] has been "decomposed," multiple right-hand-side vectors can be evaluated in an efficient manner.

L U Decomposition Method

- For most non-singular matrix [A] that one could conduct Naïve Gauss Elimination forward elimination steps, one can always write it as

  [A] = [L][U]

  where

  [L] = lower triangular matrix

  [U] =upper triangular matrix

Solve Linear System Using LUD

- Given [A][X] = [C]

1. Decompose $[A]$ into $[L]$ and $[U]$:

$$[A] = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ \ell_{21} & 1 & \cdots & 0 \\ \vdots & \vdots & \cdots & \vdots \\ \ell_{n1} & \ell_{n2} & \cdots & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & \cdots & u_{1n} \\ 0 & u_{22} & \cdots & u_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & \cdots & u_{nn} \end{bmatrix}$$

The elements of the $[U]$ matrix are exactly the same as the coefficient matrix one obtains at the end of the *forward elimination* steps in Naïve Gauss elimination.

Determination of L and U (cont.)

- The lower triangular matrix [L] *has 1 in its diagonal entries.*
- *The non-zero* elements on the non-diagonal elements in [L] *are multipliers that made the* corresponding entries zero in the upper triangular matrix [U] *during forward* elimination.

L U Decomposition Method

- If solving a set of linear equations [A][X] = [C]
- If [A] = [L][U] then [L][U][X] = [C]
- Multiply by $[L]^{-1}$
- Which gives $[L]^{-1}[L][U][X] = [L]^{-1}[C]$
- Remember $[L]^{-1}[L] = [I]$
  which leads to $[I][U][X] = [L]^{-1}[C]$
- Now, if [I][U] = [U] then $[U][X] = [L]^{-1}[C]$
- Now, let $[L]^{-1}[C]=[Z]$
- Which ends with [L][Z] = [C] (1)
- And [U][X] = [Z] (2)

Solve Linear System Using LUD

1. Determine L and U.
2. Confirm **[L][U] =[A].**
3. Solve **[L][Z] = [C] for [Z].**
4. Solve **[U][X] = [Z] for [X].**

Example

Use the LU decomposition method to solve the following simultaneous linear equations.

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

Solution

1. Decompose [*A*] into [*L*] and [*U*]:

$$[A] = [L][U] = \begin{bmatrix} 1 & 0 & 0 \\ \ell_{21} & 1 & 0 \\ \ell_{31} & \ell_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

To find l_{21} and l_{31}, find the multiplier that was used to make the a_{21} and a_{31} elements zero in the first step of forward elimination of the Naïve Gauss elimination method. It was

$$l_{21} = \frac{a_{21}}{a_{11}} = \frac{64}{25} = 2.56 \qquad l_{31} = \frac{a_{31}}{a_{11}} = \frac{144}{25} = 5.76$$

To find l_{32}, what multiplier was used to make a_{32} a element zero? Remember a_{32} a element was made zero in the second step of forward elimination.

The matrix [A] at the beginning of the second step of forward elimination was

$$\begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & -16.8 & -4.76 \end{bmatrix} \Rightarrow l_{32} = \frac{a_{32}}{a_{22}} = \frac{-16.8}{-4.8} = 3.5$$

So [L]

$$[L] = \begin{bmatrix} 1 & 0 & 0 \\ 2.56 & 1 & 0 \\ 5.76 & 3.5 & 1 \end{bmatrix}$$

The matrix [U] is the same as found at the end of the forward elimination of Naïve Gauss elimination method, that is

$$[U] = \begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & 0 & 0.7 \end{bmatrix}$$

2. Confirm

$$[L][U]=[A]$$

$$[L][U]=\begin{bmatrix}1 & 0 & 0\\ 2.56 & 1 & 0\\ 5.76 & 3.5 & 1\end{bmatrix}\begin{bmatrix}25 & 5 & 1\\ 0 & -4.8 & -1.56\\ 0 & 0 & 0.7\end{bmatrix}=\begin{bmatrix}25 & 5 & 1\\ 64 & 8 & 1\\ 144 & 12 & 1\end{bmatrix}$$

3. Solve

$$[L][Z]=[C]$$

$$\begin{bmatrix}1 & 0 & 0\\ 2.56 & 1 & 0\\ 5.76 & 3.5 & 1\end{bmatrix}\begin{bmatrix}z_1\\ z_2\\ z_3\end{bmatrix}=\begin{bmatrix}106.8\\ 177.2\\ 279.2\end{bmatrix}$$

to give

$$\begin{aligned} z_1 &= 106.8\\ z_2 &= -96.208\\ z_3 &= 0.76\end{aligned} \quad \Rightarrow \quad [Z]=\begin{bmatrix}z_1\\ z_2\\ z_3\end{bmatrix}=\begin{bmatrix}106.8\\ -96.208\\ 0.76\end{bmatrix}$$

4. Now solve $[U][X]=[Z]$

$$\begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & 0 & 0.7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ -96.208 \\ 0.76 \end{bmatrix}$$



$$x_3 = \frac{0.76}{0.7} = 1.0857$$

$$x_2 = 19.691$$

$$x_1 = 0.29048$$

Hence the solution vector is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0.29048 \\ 19.691 \\ 1.0857 \end{bmatrix}$$

Example

- Solve the system

$$\begin{bmatrix} 3 & -0.1 & -0.2 \\ 0.1 & 7 & -0.3 \\ 0.3 & -0.2 & 10 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 7.85 \\ -19.3 \\ 71.4 \end{bmatrix}$$

$$\begin{bmatrix} 3 & -0.1 & -0.2 \\ 0.1 & 7 & -0.3 \\ 0.3 & -0.2 & 10 \end{bmatrix}$$

L_{21} $\frac{0.1}{3} = 0.0333$

L_{31} $\frac{0.3}{3} = 0.1$

$$\overbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0.0333333 & 1 & 0 \\ 0.100000 & -0.0271300 & 1 \end{bmatrix}}^{\mathrm{L}}$$

First Forward Elimination: Eliminate X1

$$\begin{bmatrix} 3 & -0.1 & -0.2 \\ 0 & 7.0033 & -0.293333 \\ 0 & -0.19 & 10.02 \end{bmatrix}$$

ℓ_{32} $\frac{-0.19}{7.0033} = -0.02713$

Second Forward Elimination: Eliminate X2

$$[U] = \begin{bmatrix} 3 & -0.1 & -0.2 \\ 0 & 7.0033 & -0.293333 \\ 0 & 0 & 10.012 \end{bmatrix}$$

2- Confirm **[L][U] =[A].**

$$\begin{bmatrix} 1 & 0 & 0 \\ 0.0333333 & 1 & 0 \\ 0.100000 & -0.0271300 & 1 \end{bmatrix} \begin{bmatrix} 3 & -0.1 & -0.2 \\ 0 & 7.0033 & -0.293333 \\ 0 & 0 & 10.012 \end{bmatrix} = \begin{bmatrix} 3 & -0.1 & -0.2 \\ 0.1 & 7 & -0.3 \\ 0.3 & -0.2 & 10 \end{bmatrix}$$

Example

3. Solve **[L][Z] = [C]** **for** **[Z].**

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0.0333333 & 1 & 0 \\ 0.100000 & -0.0271300 & 1 \end{bmatrix}}_{L} \underbrace{\begin{Bmatrix} d_1 \\ d_2 \\ d_3 \end{Bmatrix}}_{Z} = \begin{Bmatrix} 7.85 \\ -19.3 \\ 71.4 \end{Bmatrix}$$

$d_1 = 7.85$

$0.0333333d_1 + d_2 = -19.3$

$0.1d_1 - 0.02713d_2 + d_3 = 71.4$

4. Solve [U][X] = [Z] for [X].

- $[Z] = \begin{bmatrix} 7.85 \\ -19.5617 \\ 70.0843 \end{bmatrix}$
- [U][X]=[Z]
- $[X] = \begin{bmatrix} 3 \\ -2.3 \\ 7.00003 \end{bmatrix}$

[U]= $\begin{bmatrix} 3 & -0.1 & -0.2 \\ 0 & 7.0033 & -0.293333 \\ 0 & 0 & 10.012 \end{bmatrix}$

4- Gauss Seidel Method

Gauss Seidel

- Iterative or approximate methods provide an alternative to the elimination methods described to this point. Such approaches are similar to the techniques we developed to obtain the roots of a single equation.

Given a general set of n equations and n unknowns, we have

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \ldots + a_{1n}x_n = c_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \ldots + a_{2n}x_n = c_2$$

$$\vdots \qquad\qquad \vdots$$

$$a_{n1}x_1 + a_{n2}x_2 + a_{n3}x_3 + \ldots + a_{nn}x_n = c_n$$

$$x_2 = \frac{c_2 - a_{21}x_1 - a_{23}x_3 \ldots\ldots - a_{2n}x_n}{a_{22}}$$

$$\vdots$$

$$\vdots$$

$$x_{n-1} = \frac{c_{n-1} - a_{n-1,1}x_1 - a_{n-1,2}x_2 \ldots\ldots - a_{n-1,n-2}x_{n-2} - a_{n-1,n}x_n}{a_{n-1,n-1}}$$

$$x_n = \frac{c_n - a_{n1}x_1 - a_{n2}x_2 - \ldots\ldots - a_{n,n-1}x_{n-1}}{a_{nn}}$$

$$x_1 = \frac{c_1 - \sum_{\substack{j=1 \\ j \neq 1}}^{n} a_{1j}x_j}{a_{11}}$$

$$x_2 = \frac{c_2 - \sum_{\substack{j=1 \\ j \neq 2}}^{n} a_{2j} x_j}{a_{22}}$$

.

.

.

$$x_{n-1} = \frac{c_{n-1} - \sum_{\substack{j=1 \\ j \neq n-1}}^{n} a_{n-1,j} x_j}{a_{n-1,n-1}}$$

$$x_n = \frac{c_n - \sum_{\substack{j=1 \\ j \neq n}}^{n} a_{nj} x_j}{a_{nn}}$$

$$\left| \in_a \right|_i = \left| \frac{x_i^{\text{new}} - x_i^{\text{old}}}{x_i^{\text{new}}} \right| \times 100$$

Gauss Seidel

- Now to find x_i, one assumes an initial guess for the x_i and then uses the rewritten equations to calculate the new estimates.
- Remember, one always uses the most recent estimates to calculate the next estimates x_i .
- At the end of each iteration, one calculates the absolute relative approximate error for each x_i as

$$\left|\epsilon_a\right|_i = \left|\frac{x_i^{\text{new}} - x_i^{\text{old}}}{x_i^{\text{new}}}\right| \times 100$$

where is the recently obtained value of x_i , and x_i^{old} is the previous value of x_i .

- When the absolute relative approximate error for each x_i is less than the pre-specified tolerance, the iterations are stopped.

Convergence

- If coefficient matrix of the linear equations forms a ***diagonally dominant*** matrix, Hence the solution will converge. If not the solution ***might not*** converge.

- Diagonally dominant $\Rightarrow$ $|a_{ii}| > \sum_{\substack{j=1 \\ j \neq i}}^{n} |a_{ij}|$

This criterion is sufficient but not necessary for convergence

Gauss-Seidel Method

Problem Statement. Use the Gauss-Seidel method to obtain the solution of the same system used in Example 10.2:

$$\begin{aligned} 3x_1 - 0.1x_2 - 0.2x_3 &= 7.85 \\ 0.1x_1 + 7x_2 - 0.3x_3 &= -19.3 \\ 0.3x_1 - 0.2x_2 + 10x_3 &= 71.4 \end{aligned}$$

Recall that the true solution is $x_1 = 3$, $x_2 = -2.5$, and $x_3 = 7$.

Solution. First, solve each of the equations for its unknown on the diagonal.

$$x_1 = \frac{7.85 + 0.1x_2 + 0.2x_3}{3} \tag{E11.3.1}$$

$$x_2 = \frac{-19.3 - 0.1x_1 + 0.3x_3}{7} \tag{E11.3.2}$$

$$x_3 = \frac{71.4 - 0.3x_1 + 0.2x_2}{10} \tag{E11.3.3}$$

By assuming that x_2 and x_3 are zero, Eq. (E11.3.1) can be used to compute

$$x_1 = \frac{7.85 + 0 + 0}{3} = 2.616667$$

This value, along with the assumed value of $x_3 = 0$, can be substituted into Eq. (E11.3.2) to calculate

$$x_2 = \frac{-19.3 - 0.1(2.616667) + 0}{7} = -2.794524$$

The first iteration is completed by substituting the calculated values for x_1 and x_2 into Eq. (E11.3.3) to yield

$$x_3 = \frac{71.4 - 0.3(2.616667) + 0.2(-2.794524)}{10} = 7.005610$$

For the second iteration, the same process is repeated to compute

$$x_1 = \frac{7.85 + 0.1(-2.794524) + 0.2(7.005610)}{3} = 2.990557 \qquad |\varepsilon_t| = 0.31\%$$

$$x_2 = \frac{-19.3 - 0.1(2.990557) + 0.3(7.005610)}{7} = -2.499625 \qquad |\varepsilon_t| = 0.015\%$$

$$x_3 = \frac{71.4 - 0.3(2.990557) + 0.2(-2.499625)}{10} = 7.000291 \qquad |\varepsilon_t| = 0.0042\%$$

The method is, therefore, converging on the true solution. Additional iterations could be applied to improve the answers. However, in an actual problem, we would not know the true answer a priori. Consequently, Eq. (11.6) provides a means to estimate the error. For example, for x_1,

$$|\varepsilon_{a,1}| = \left|\frac{2.990557 - 2.616667}{2.990557}\right| 100\% = 12.5\%$$

For x_2 and x_3, the error estimates are $|\varepsilon_{a,2}| = 11.8\%$ and $|\varepsilon_{a,3}| = 0.076\%$. Note that, as was the case when determining roots of a single equation, formulations such as Eq. (11.6) usually provide a conservative appraisal of convergence. Thus, when they are met, they ensure that the result is known to at least the tolerance specified by ε_s.

Example

Find the solution to the following system of equations using the Gauss-Seidel method.

$$21x_1 + 3x_2 - 5x_3 = 1$$

$$x_1 + 5x_2 + 3x_3 = 28$$

$$3x_1 + 7x_2 + 13x_3 = 76$$

Use

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

as the initial guess and conduct two iterations.

Solution

The coefficient matrix

$$[A] = \begin{bmatrix} 12 & 3 & -5 \\ 1 & 5 & 3 \\ 3 & 7 & 13 \end{bmatrix}$$

is diagonally dominant as

$$|a_{11}| = |12| = 12 \geq |a_{12}| + |a_{13}| = |3| + |-5| = 8$$

$$|a_{22}| = |5| = 5 \geq |a_{21}| + |a_{23}| = |1| + |3| = 4$$

$$|a_{33}| = |13| = 13 \geq |a_{31}| + |a_{32}| = |3| + |7| = 10$$

and the inequality is strictly greater than for at least one row. Hence, the solution should converge using the Gauss-Seidel method.

Rewriting the equations, we get

$$x_1 = \frac{1 - 3x_2 + 5x_3}{12}$$

$$x_2 = \frac{28 - x_1 - 3x_3}{5}$$

$$x_3 = \frac{76 - 3x_1 - 7x_2}{13}$$

Assuming an initial guess of

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

- **Iteration #1**

$$x_1 = \frac{1-3(0)+5(1)}{12} = 0.5000$$

$$x_2 = \frac{28-(0.5000)-3(1)}{5} = 4.9000$$

$$x_3 = \frac{76-3(0.5000)-7(4.9000)}{13} = 3.0923$$

The absolute relative approximate error at the end of the first iteration is

$$|\epsilon_a|_1 = \left|\frac{0.5000-1}{0.5000}\right| \times 100 = 100\%$$

$$|\epsilon_a|_2 = \left|\frac{4.9000-0}{4.9000}\right| \times 100 = 100\%$$

$$|\epsilon_a|_3 = \left|\frac{3.0923-1}{3.0923}\right| \times 100 = 67.662\%$$

The maximum absolute relative approximate error is 100.00%

Iteration #2

$$x_1 = \frac{1 - 3(4.9000) + 5(3.0923)}{12} = 0.14679$$

$$x_2 = \frac{28 - (0.14679) - 3(3.0923)}{5} = 3.7153$$

$$x_3 = \frac{76 - 3(0.14679) - 7(3.7153)}{13} = 3.8118$$

At the end of second iteration, the absolute relative approximate error is

$$|\epsilon_a|_1 = \left|\frac{0.14679 - 0.50000}{0.14679}\right| \times 100 = 240.61\%$$

$$|\epsilon_a|_2 = \left|\frac{3.7153 - 4.9000}{3.7153}\right| \times 100 = 31.889\%$$

$$|\epsilon_a|_3 = \left|\frac{3.8118 - 3.0923}{3.8118}\right| \times 100 = 18.874\%$$

The maximum absolute relative approximate error is 240.61%. This is greater than the value of 100.00% we obtained in the first iteration. Is the solution diverging? No, as you conduct more iterations, the solution converges as follows

Iteration	x_1	$\lvert\epsilon_a\rvert_1\%$	x_2	$\lvert\epsilon_a\rvert_2\%$	x_3	$\lvert\epsilon_a\rvert_3\%$
1	0.50000	100.00	4.9000	100.00	3.0923	67.662
2	0.14679	240.61	3.7153	31.889	3.8118	18.874
3	0.74275	80.236	3.1644	17.408	3.9708	4.0064
4	0.94675	21.546	3.0281	4.4996	3.9971	0.65772
5	0.99177	4.5391	3.0034	0.82499	4.0001	0.074383
6	0.99919	0.74307	3.0001	0.10856	4.0001	0.00101

This is close to the exact solution vector of

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}$$

Example

Given the system of equations

$$3x_1 + 7x_2 + 13x_3 = 76$$

$$x_1 + 5x_2 + 3x_3 = 28$$

$$12x_1 + 3x_2 - 5x_3 = 1$$

find the solution using the Gauss-Seidel method. Use

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

Solution

Rewriting the equations, we get

$$x_1 = \frac{76 - 7x_2 - 13x_3}{3}$$

$$x_2 = \frac{28 - x_1 - 3x_3}{5}$$

$$x_3 = \frac{1 - 12x_1 - 3x_2}{-5}$$

Assuming an initial guess of

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

the next six iterative values are given in the table below.

Iteration	x_1	$\lvert\epsilon_a\rvert_1\%$	x_2	$\lvert\epsilon_a\rvert_2\%$	x_3	$\lvert\epsilon_a\rvert_3\%$
1	21.000	95.238	0.80000	100.00	50.680	98.027
2	–196.15	110.71	14.421	94.453	–462.30	110.96
3	1995.0	109.83	–116.02	112.43	4718.1	109.80
4	–20149	109.90	1204.6	109.63	–47636	109.90
5	2.0364×10^5	109.89	–12140	109.92	4.8144×10^5	109.89
6	-2.0579×10^6	109.89	1.2272×10^5	109.89	-4.8653×10^6	109.89

You can see that this solution is not converging and the coefficient matrix is not diagonally dominant. The coefficient matrix

$$[A]=\begin{bmatrix}3 & 7 & 13\\ 1 & 5 & 3\\ 12 & 3 & -5\end{bmatrix}$$

is not diagonally dominant as

$$|a_{11}|=|3|=3\le|a_{12}|+|a_{13}|=|7|+|13|=20$$